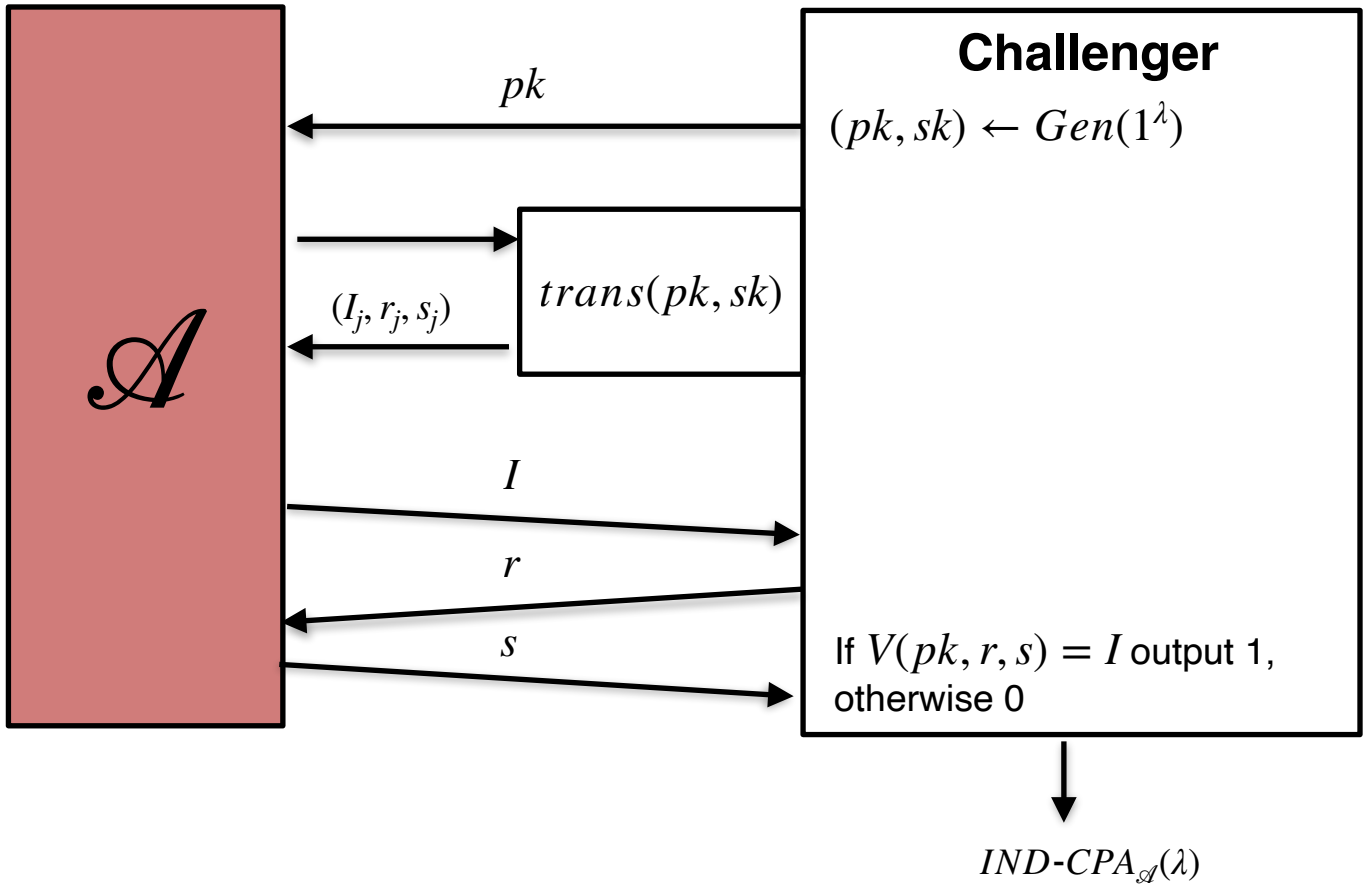
$Ident_{\mathscr{A}}(\lambda)$ Experiment:



Definition

An encryption scheme (Gen, P_1, P_2, V) is **secure**, if it holds for **every PPT-adversary** $\mathscr{A}$ there exists a negligible function ν s.t. for all $\lambda \in \mathbb{N}$

$$\Pr[Ident_{\mathscr{A}}(\lambda) = 1] < \frac{1}{2} + \nu(\lambda)$$

Recap: The Discrete Logarithm Problem

- Let $\mathbb{G}$ be a cryptographic group of order p with generator g
- We omit group generation algorithm, but bear in mind that this is actually an infinite family of groups
- The discrete logarithm assumption in $\mathbb{G}$ conjectures that it holds for every PPT algorithm $\mathscr{A}$ that
  $\Pr[\mathscr{A}(g^x) = x] \leq negl(\lambda)$
  where the probability is taken over the random choice of $x \leftarrow_{\$} \mathbb{Z}_p$

Schnorr's Identification Scheme

$$x \leftarrow_{\$} \mathbb{Z}_p \quad y \leftarrow g^x$$
$$sk \leftarrow x \quad pk \leftarrow y$$

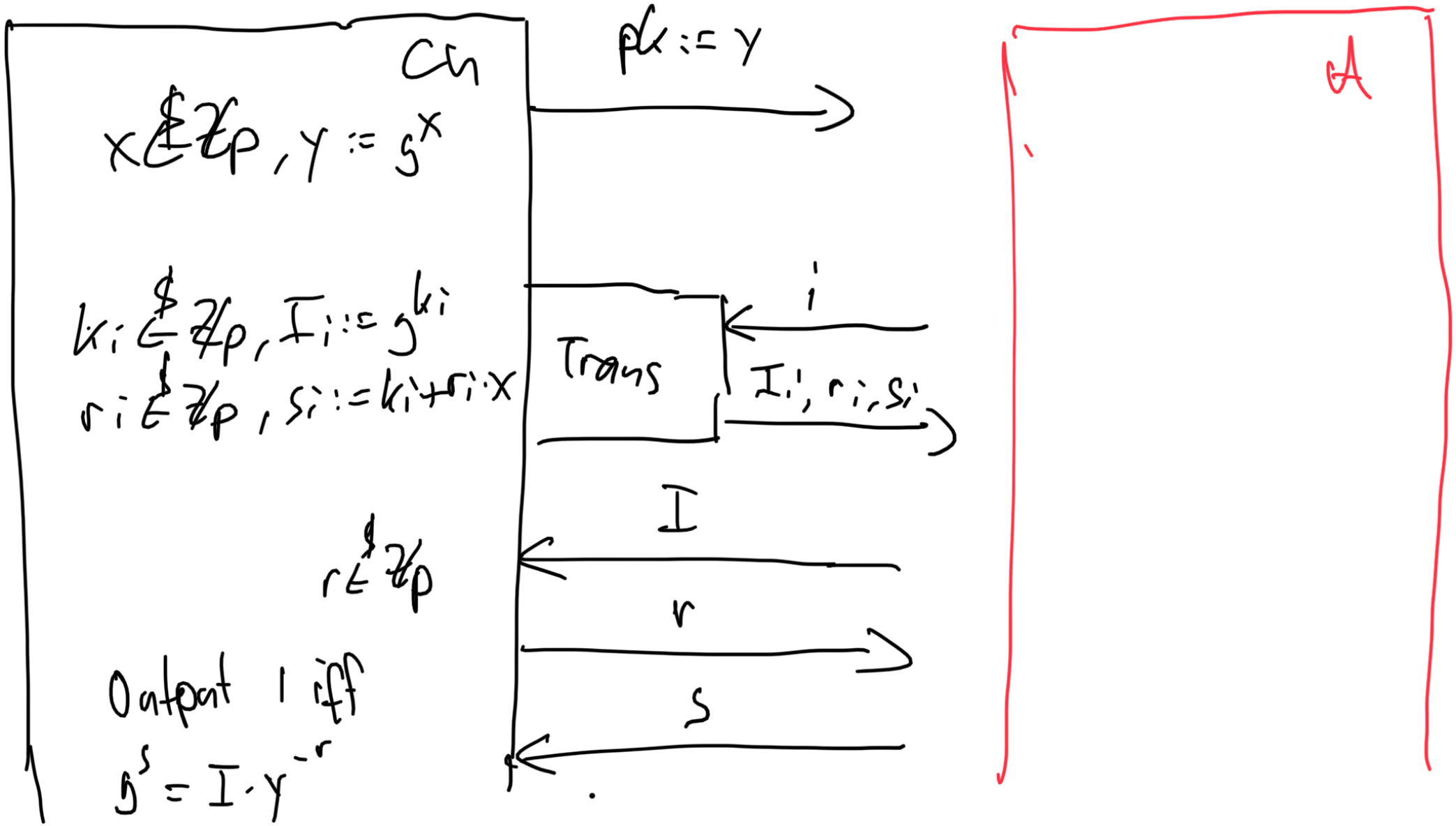
Prover(sk)		Verifier(pk)
$k \leftarrow_{\$} \mathbb{Z}_p$ $I \leftarrow g^k$	$\xrightarrow{I}$	
		$r \leftarrow_{\$} \mathbb{Z}_p$
	$\xleftarrow{r}$	
$s \leftarrow r \cdot x + k \mod p$	$\xrightarrow{s}$	
		Accept if $g^s \cdot y^{-r} = I$

Security

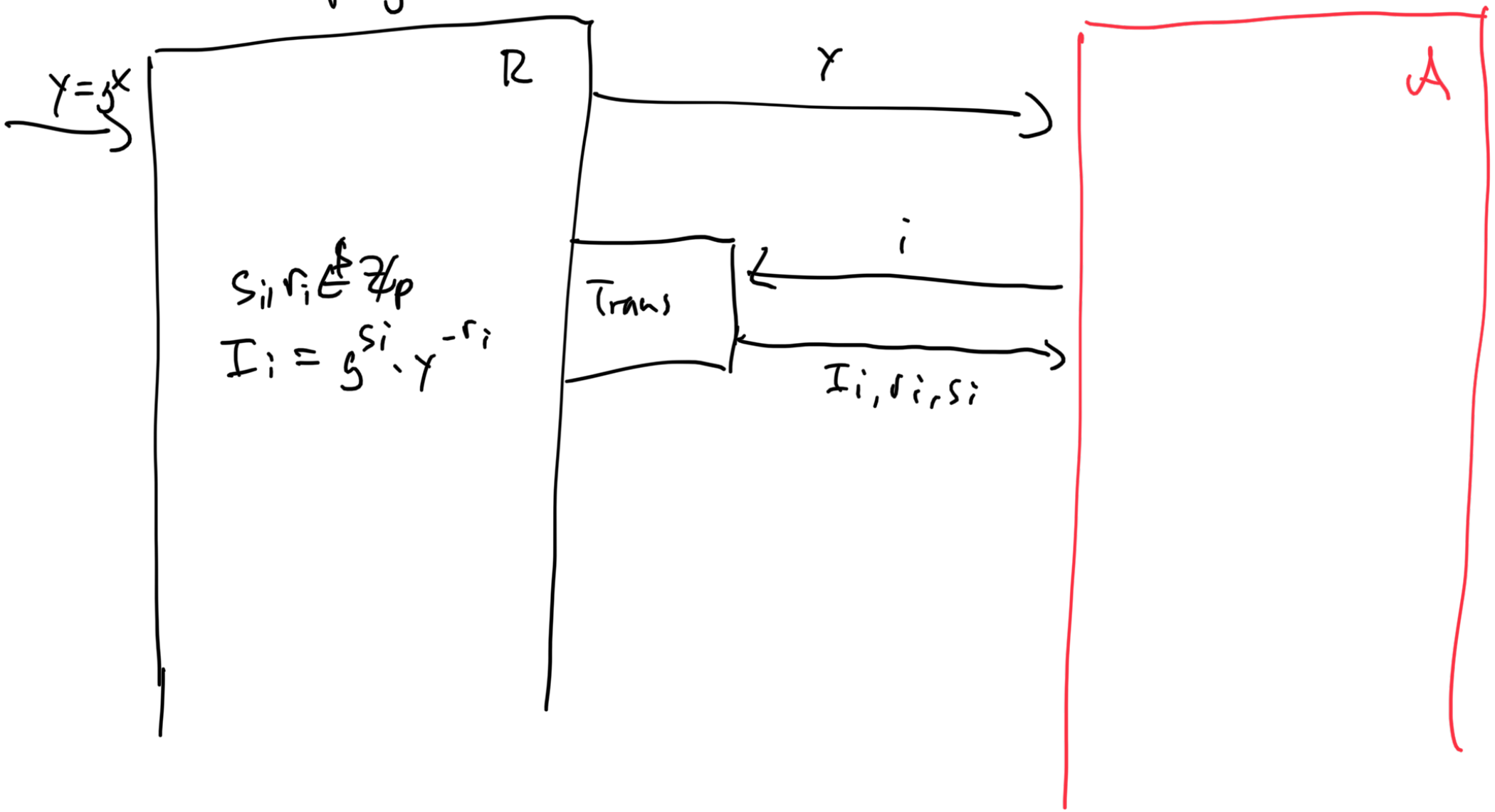
Theorem 9.3

- Assume that the discrete logarithm assumption holds in the group $\mathbb{G}$
- Then the Schnorr Identification Scheme is secure

Proof


Ch
$x \xleftarrow{\$} \mathbb{Z}_p, y := g^x$
$pk := y$
A
$k_i \xleftarrow{\$} \mathbb{Z}_p, I_i := g^{k_i}$
$r_i \xleftarrow{\$} \mathbb{Z}_p, s_i := k_i + r_i \cdot x$
Trans
i
I_i, r_i, s_i
I
$r \xleftarrow{\$} \mathbb{Z}_p$
r
s
Output 1 iff
$g^s = I \cdot y^{-r}$

Proof Simulating Transcripts:



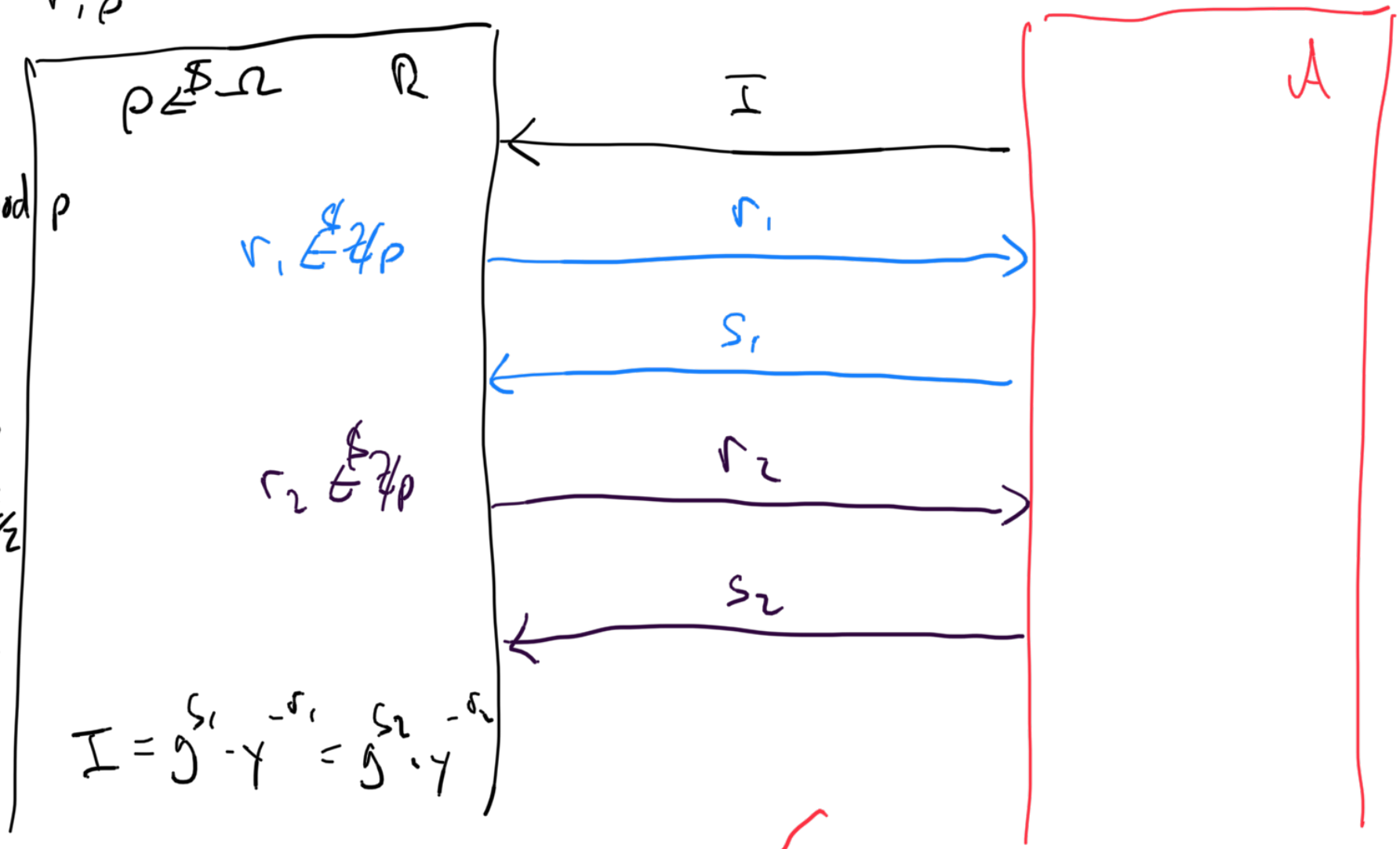
Proof

$\Pr_{r,\rho}[\mathrm{Ident}^{\mathcal{A}} = 1] \geq \varepsilon$

$x = \frac{s_1 - s_2}{(r_1 - r_2)^{-1}} \bmod p$

$(i): (\rho, r_1) \in B$ with prob $\geq \varepsilon/2$ and $\mathcal{A}$ wins the first time

By properties of B, $\mathcal{A}$ wins a second time with probability at least $\varepsilon/2$

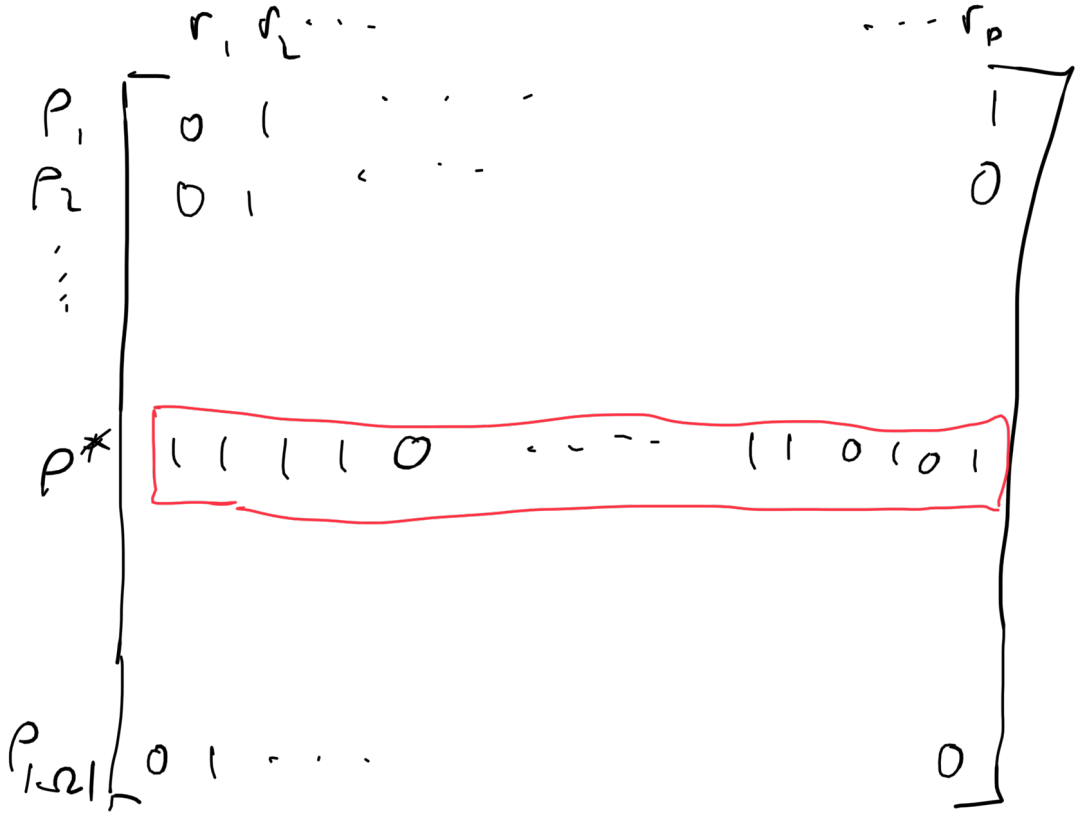


$\frac{\varepsilon^2}{4} - \frac{1}{p}$

The Heavy Row Lemma

Theorem 9.3

- Suppose that $\Pr_{r,\rho}[Ident_{\mathscr{A}}^{r,\rho} = 1] \geq \epsilon$
- Define $B := \{(r,\rho) \mid \Pr_{r \leftarrow \mathbb{Z}_p}[Ident_{\mathscr{A}}^{r,\rho} = 1] \geq \epsilon/2\}$
- Then:
  - $\Pr_{r,\rho}[(r,\rho) \in B \mid Ident_{\mathscr{A}}^{r,\rho} = 1\} \geq \frac{1}{2}$
  - $\Pr_{r,\rho}[(r,\rho) \in B\} \geq \frac{\epsilon}{2}$



Proof

(ii) $\Pr_{r,\rho}[(r,\rho) \in B] = \Pr_{r,\rho}[(r,\rho) \in B \mid \mathcal{A} \text{ wins on } r,\rho] \cdot \Pr_{r,\rho}[\mathcal{A} \text{ wins}]$

$+ \Pr_{r,\rho}[(r,\rho) \in B \mid \mathcal{A} \text{ loses on } (r,\rho)] \cdot \Pr_{r,\rho}[\mathcal{A} \text{ loses on } r,\rho]$

$\geq \Pr_{r,\rho}[(r,\rho) \in B \mid \mathcal{A} \text{ wins on } r,\rho] \cdot \underbrace{\Pr_{r,\rho}[\mathcal{A} \text{ wins on } r,\rho]}_{\geq \varepsilon}$

$\overset{(i)}{\geq} \frac{1}{2} \cdot \varepsilon$

(i) $\Pr_{r,\rho}[(r,\rho) \in B \mid \mathcal{A} \text{ wins on } r,\rho] = 1 - \Pr_{r,\rho}[(r,\rho) \notin B \mid \mathcal{A} \text{ wins on } (r,\rho)]$

$$= 1 - \frac{\Pr_{r,\rho}[\mathcal{A} \text{ wins on } (r,\rho) \mid (r,\rho) \notin B] \cdot \Pr_{r,\rho}[(r,\rho) \notin B]}{\Pr_{r,\rho}[\mathcal{A} \text{ wins on } (r,\rho)]}$$

Proof

$$\geq 1 - \Pr_{r,\rho}[\mathcal{A} \text{ wins on } (r,\rho) \mid (r,\rho) \notin B] \cdot 1/\varepsilon$$

$$\geq 1 - \frac{1}{\varepsilon} \cdot \sum_{\hat{\rho} \in \Omega} \Pr_{r,\rho}[\mathcal{A} \text{ wins on } (r,\rho) \mid (\rho = \hat{\rho}) \wedge (r,\rho) \notin B]$$

$$\cdot \Pr_{r,\rho}[\rho = \hat{\rho} \mid (r,\rho) \notin B]$$

$$= 1 - \frac{1}{\varepsilon} \cdot \sum_{\hat{\rho} \in \Omega} \Pr_{r}[\mathcal{A} \text{ wins on } (r,\hat{\rho}) \mid (r,\hat{\rho}) \notin B] \cdot \Pr_{r,\rho}[\rho = \hat{\rho} \mid (r,\rho) \notin B]$$

$$> 1 - \frac{1}{\varepsilon} \sum_{\hat{\rho} \in \Omega} \varepsilon/2 \cdot \Pr_{r,\rho}[\rho = \hat{\rho} \mid (r,\rho) \notin B] = 1 - \varepsilon/2 \cdot \frac{1}{\varepsilon} \cdot \underbrace{\sum_{\hat{\rho} \in \Omega} \Pr_{r,\rho}[\rho = \hat{\rho} \mid (r,\rho) \notin B]}_{=1}$$

$$= 1 - \frac{1}{2} = \frac{1}{2} \quad \square$$

Schnorr Signatures

- Schnorr signatures are basically the most efficient and shortest signatures when instantiated with elliptic curve groups
- But they are not widely used!
- In practice the group $\mathbb{G}$, its order p and a generator g are publicly fixed, e.g Ed25519 https://ed25519.cr.yp.to
- Size of verification and signing keys: 256 bits (32 bytes)
- Size of signatures 512 bits (64 bytes)
- Signing: 1 exponentiation
- Verifying: 2 exponentiations
- Intellectual Property Rights!
- https://patents.google.com/patent/US4995082

The Digital Signature Algorithm (DSA)

- Let $\mathbb{G}$ be a cryptographic group of order p with generator g
- Let $F, H : \{0,1\}^* \to \mathbb{Z}_p$ be two hash functions
- The DSA scheme $(Gen, Sign, Verify)$ is given as follows
  - $Gen(1^\lambda)$: Choose $x \leftarrow_\$ \mathbb{Z}_p$, compute $y \leftarrow_\$ g^x$, output $vk \leftarrow y$ and $sk \leftarrow x$
  - $Sign(sk = x, m)$: Choose $k \leftarrow_\$ \mathbb{Z}_p$, compute $r \leftarrow F(g^k)$, $s \leftarrow k^{-1} \cdot (H(m) + xr) \mod p$, restart if $r = 0$ or $s = 0$. Output $\sigma \leftarrow (r, s)$
  - $Verify(vk = y, m, \sigma = (r, s))$: If $r, s \neq 0$ and $r \leftarrow F(g^{H(m)s^{-1}} y^{rs^{-1}})$, if $r = H(I, m)$ output 1, otherwise 0

Bilinear Pairings

- A (symmetric) **bilinear pairing** is a mapping $e : \mathbb{G} \times \mathbb{G} \longrightarrow \mathbb{G}_T$ with the following properties:
  - **Non-Degeneracy:** $e(g, g) \neq 1_T$
  - **Symmetry:** $e(g^a, g^b) = e(g^b, g^a) = e(g, g)^{ab}$
  - **Bilinearity:** $e(g^a, g) \cdot e(g, g^b) = e(g^{a+b}, g)$
- $\mathbb{G}$ is usually an elliptic curve group, $\mathbb{G}_T$ is usually a subgroup of $\mathbb{F}_q$
- **Key Property:** Pairings give a Decision Diffie-Hellman oracle over $\mathbb{G}$
- Given g^a, g^b, g^c, one can efficiently check whether $e(g^a, g^b) = e(g, g)^{ab} = e(g^c, g)$

The BLS Signature Scheme

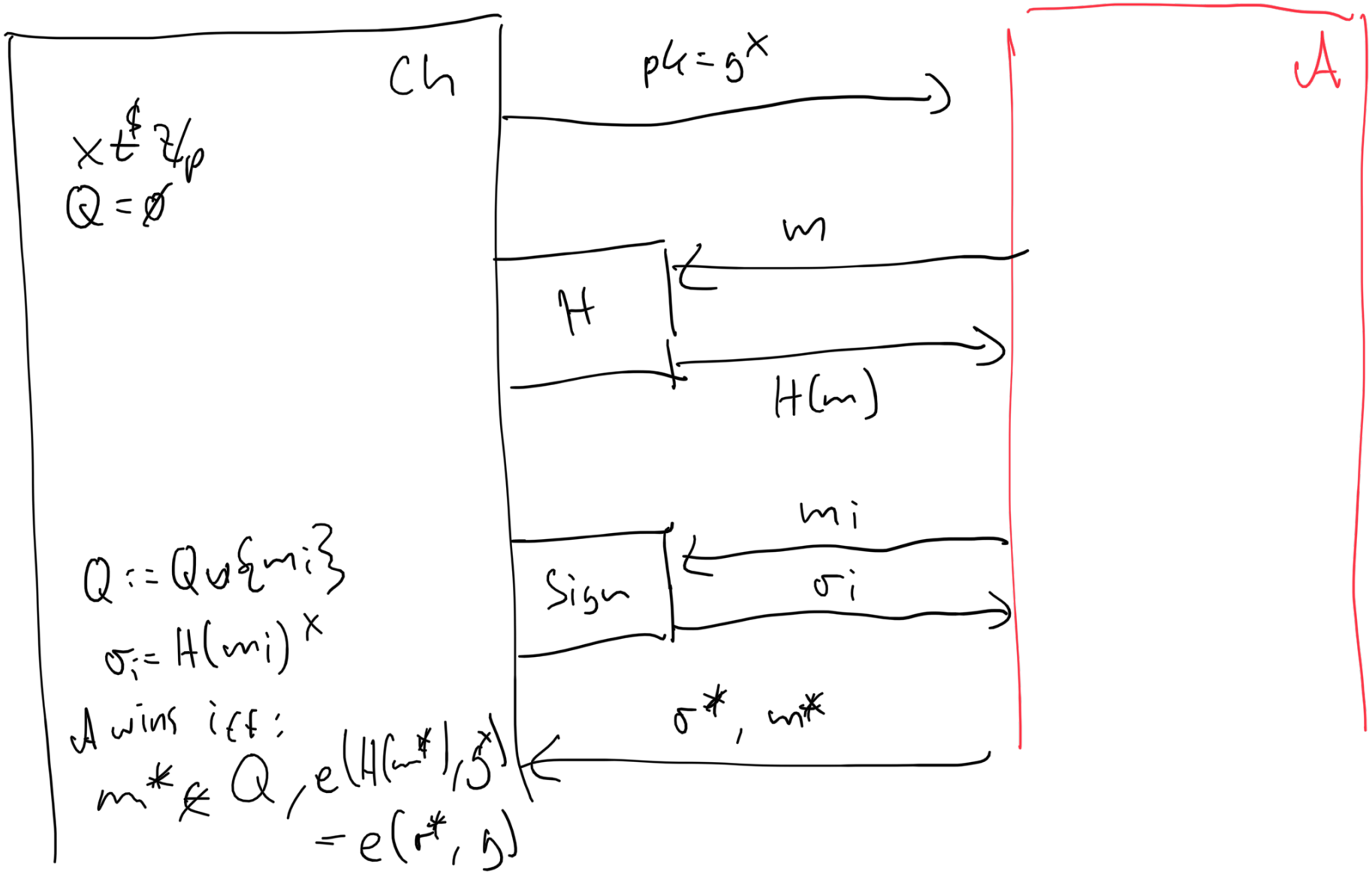
- Invented by Boneh, Lynn, and Schacham in 2001
- Hash-and-Sign type Signature
- The BLS Hash-and-Sign scheme $(Gen, Sign, Verify)$ is given as follows
  - $Gen(1^\lambda)$: Sample $x \leftarrow \mathbb{Z}_p$. Output $vk \leftarrow g^x$ and $sk \leftarrow x$
  - $Sign(x, m)$: Compute and output $\sigma \leftarrow H(m)^x \in \mathbb{G}$
  - $Verify(vk = g^x, m, \sigma)$: If $e(H(m), g^x) = e(g, \sigma)$ output 1, otherwise 0
- **Pro:** Yields very short signatures
- **Con:** Pairings are expensive

Security

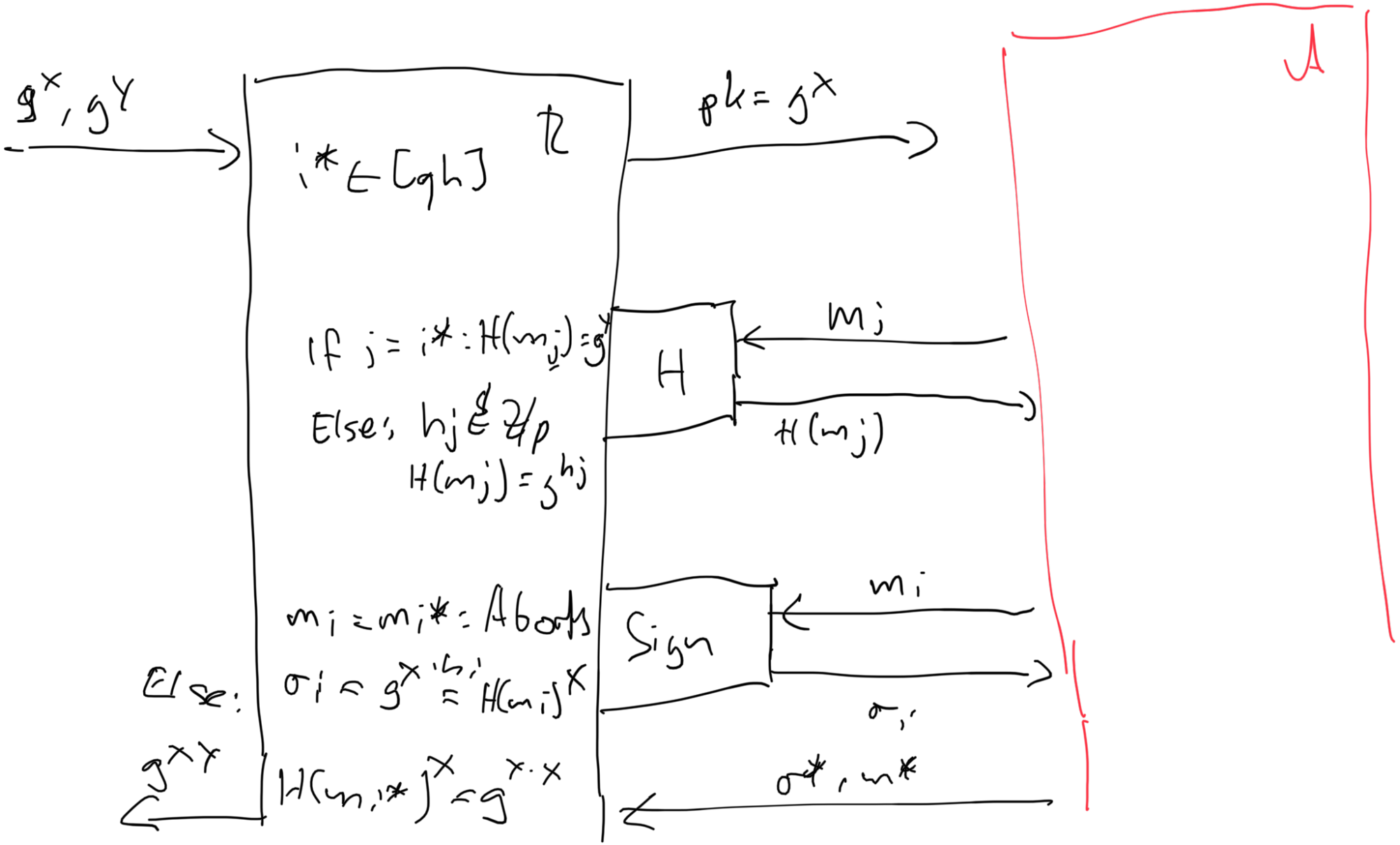
Theorem 9.4

- Assume that the CDH assumption holds in the group $\mathbb{G}$
- Then the BLS Scheme is secure

Proof


Ch
A
$x \xleftarrow{\$} \mathbb{Z}_p$
$Q = \emptyset$
$pk = g^x$
m
H
$H(m)$
m_i
Sign
σ_i
$Q := Q \cup \{m_i\}$
$\sigma_i = H(m_i)^x$
σ^*, m^*
A wins iff:
$m^* \notin Q, e(H(m^*), g^x) = e(\sigma^*, g)$

Proof


g^x, g^y
$i^* \leftarrow [q_h]$
R
$pk = g^x$
A
If $j = i^*$: $H(m_j) = g^y$
Else: $h_j \xleftarrow{\$} \mathbb{Z}_p$
$H(m_j) = g^{h_j}$
H
m_j
$H(m_j)$
$m_i = m_{i^*}$: Abort
$\sigma_i = g^{x \cdot h_i} = H(m_i)^x$
Sign
m_i
σ_i
Else:
$H(m_{i^*})^x = g^{x \cdot y}$
g^{xy}
σ^*, m^*

Proof

Summary

- Naive attempts for access control are often susceptible to replay attacks
- Identification schemes provide security against replay attacks
- The Schnorr identification scheme is secure under the discrete logarithm assumption
- Using the Fiat-Shamir transform, the Schnorr identification scheme yields the currently most efficient know signature scheme with the shortest signatures
- Due to patent issues the slightly more contrived DSA scheme was standardized
- Bilinear Pairings offer alternative design paradigm for signatures over elliptic curve groups
- BLS signature is the shortest known signature and is secure under CDH